

ActInf Textbook Group ~ Cohort 2 ~ Meeting 3 (Chapter 1, part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_2 | Cohort 2 ~ Meeting 3 (Chapter 1, part 2) | 57:51

Okay, well, welcome everyone. Thanks for joining. It's September 16th and we are in the third meeting for Cohort 2. We're on our second discussion of Chapter 1 in the textbook. So this is our closing discussion on Chapter 1, which is going to be just our jumping off point into the following chapters. We'll go to the questions page where I saw some questions were submitted. We can also look at other areas, like just the chapter notes more generally. But before we go to the questions, is there any just remarks that people would like to share on Chapter 1? And then otherwise, I'll make one short note about another resource.

I'll be watching the hand-raising tip. Well, first, maybe I'll just share this other resource.

And it's already added to the resources section and I'll put it in the nearby.

It's called Pools of Insight. It's from 1999. However, the author is still around. And when I emailed them, yeah, this is the most recent version. And the document is called Pools of Insight.

A previous version, I guess, was called Knowledge Hydrant. But again, this is the most recent version, even though it's almost as old as some of us might be. And it's based around Alexander's, Christopher Alexander's pattern language, which may be familiar to some. It's a very unique book.

And it speaks to archetypes in architecture and design and talks about how there's composition patterns and mix of language and grammar for that. And that kind of a pattern language approach has been built on in many other settings. And this work is a very interesting application of pattern languages to learning groups. I think going through it could be interesting for anybody who's just curious about this approach of pattern languages and learning groups. And also, we're very interested to apply this more intentionally as we continue to iterate on the learning groups format in Active Inference Institute, textbook groups for years to come and new textbooks to come and so on.

Just looking at the titles, there's a lot of insight. Like the book is a knowledge hydrant. How do we even approach that? The group is like a pool of insight. How do we approach that? You know, how are we going to integrate our individual study with group dialogue? So not much more to say on this today, but if anyone's like excited about that, it would be awesome to hear their, how they see these patterns playing out because a lot of them speak to what we want to cultivate in this group. Okay.

Going to the questions page.

Again, we'll see we'll see we'll see we'll see we'll see we'll see we'll see we'll see

not the room. I'll add it there too. And it's also in resources. You can just look up, you know, pools and you'll find it.

People may still add more questions to the page, but we have some really rich questions provided already. So we'll start with a smaller one. In relation to perception action loops, the book states, the possibilities for control and adaptation are plentiful, but very few are useful. I'm not sure how useful is being used here. Useful for evolution, survival, what determines useful? Page four.

What does anyone see or think?

So I don't know. I would just venture to jump off the cliff and guess that maybe the authors mean like full maintenance of the non-equilibrium steady state density or like existence. Just to guess though. Right. What's the non-equilibrium steady state?

It is like the balance of all things that we find ourselves in as living creatures. So like, you know, I'm a human. I have, you know, a certain amount of cells, tissues, and organs. I have a temperature that I expect to find my body at and like a range of temperatures that I can survive in. I need a certain concentration of oxygen. My blood oxygen needs to be maintained at that level. So just like all of the things that like, I need to have a certain amount of water, like, like all of the things that contribute to me being able to persist as in my non-equilibrium steady state. So it's like, you know, if you're in equilibrium as a biological organism, you're dead. So, um, life is maintained in this kind of non-equilibrium steady state. And I think that we have to be able to maintain that. Um, and, and maybe the authors just mean maintenance useful in the maintenance of that non-equilibrium steady state.

Great. Yes. So one qualitative way to frame it is with persistence and survival. Not saying this is the only meaning of utility, but persistence and survival are imperatives for things. Again, using thing in the active inference sense as like the particular states, the blanket states and the internal states for things that are to persist, then they have to at least be acting in a survivable way. Things that are acting as if they are to persist will be repeatedly measured. They'll be repeatedly observed by others, or they'll be repeatedly observing themselves in the autopoietic case. And then Blue brought in, um, a nice term from the act in fontology, which is the non-equilibrium steady state. And by like mousing over, you can see some of the

definitions that have been provided. And this is all part of the work that people can contribute to if

they want to be improving definitions and so on. And the first definition is slightly more technical. The second definition is slightly less technical. And it's just says it's a system with dynamics that are unchanging or out of stationarity in some state. So, um, that doesn't mean the system is static or unchanging. It's like a dynamic balance. So we can think about like the length of a bone is in a dynamic balance between, um, being built up and being broken down. And so that maintenance of a non-equilibrium steady state is maintained kind of like homeostasis or like body temperature, blood sugar, et cetera, like Blue mentioned in the long run. Um, and that may take energy that may be like an exothermic process to maintain there, but it also is maintaining a stasis in some other respect. Um, so again, few are useful. What is this first part? Why are the possibilities for control and adaptation plentiful? And why does that present an issue? Ali?

Ali. I was just going to mention the, uh, three phases of Bayesian mechanics. Uh, I mean, mode matching dynamics, mode tracking dynamics and path tracking dynamics, uh, from which, uh, active inference dynamics, uh, falls in the category of the path tracking dynamics. Uh, but, uh, unlike the first two types, I mean, mode matching dynamics and mode tracking dynamics, uh, which, uh, requires density, uh,

I mean, which requires NESS assumption, uh, the path tracking dynamics. Yeah, that's it. Uh, is present, is described as a known as assumption dynamics. So, uh, if we extend the active inference framework to include,

not only, uh, not only, uh, the, uh, I mean, NESS preserving a dynamic to all the other kinds of agents, which, uh, just do this kind of path tracking dynamic, then, uh, I guess we would not need, uh, any kind of NESS preserving, uh, mechanism to do these kinds of action perception, uh, perception action loops.

Very, um, interesting. Um, this is bringing in another resource, which is an excellent contemporary resource, um, which is some of the recent work of Ramsted et al. on, on Bayesian mechanics. And it's, it's unpacked in guest stream 23.1. So that, that's one very nice resource. Um, and coming back to what Blu said about non-equilibrium steady states, we can think about one case as being like, there's a fixed point and we want just simple convergence to a fixed point.

And that might be useful. Like if utility in a situation is defined as we want the plate to be balanced on the stick and we want it, the plate to be as tall as possible. That's like kind of a classical control theory setting. Then we might want to have that one fixed mode and that's our utility.

It's not some sort of like absolute theological utility. It's just saying like within this domain or aspect or dimension, utility is going to be defined by reducing the divergence to some fixed mode. Alternatively, there could be a dynamic mode following like, um, there's going to be some cursor on your screen and you're just going to be seeking to follow it. And in that case, utility is going to be defined by the divergence minimization to a moving mode. And then the last case is this path matching, which does introduce some, um, more powerful and general mathematics, like, um, the gauge, the gauge formalism. And in that case, the path distances are being divergence minimized.

And it brings in a super important topic that, that, um, next I, I, um, will turn to Brock to give

a thought on, which is this idea of utility and pragmatic value as divergence minimization.

We're going to come back to that, but Brock, um, feel free to add what you would like about this if you want.

Um, I mean, it's page four, so it might be, there's a lot more like as we go through this, that we'll kind of start helping to connect dots. But, um, in the kind of biological, uh, homeostasis can like example, the blues, um,

there are just a lot of things that you could do that would not be, like you're saying in that context, useful for the organism. Um, and there are things that you could do that are useful maybe in a moment, but they're not kind of meta useful, like, or like creating more usefulness or something like this for the organism, right? Um, but there are, are things that, that an organism might do that like, um, just kind of put it in a place where all the things that could do are just significantly more likely to be useful. And so one of those things that it might do is like, go find forage for like a place where food is plentiful. And then if it gets hungry and it just starts to go look, then necessarily it will kind of start finding food more readily than if it hadn't gone to that patch or, you know, humans, we build houses or some sort of structure that protects us from the elements. And then, you know, um, if we oversleep or whatever, like a wild animal doesn't get us and neither does the cold or the heat maybe in the same way. Like, um, so that's, um, the language that the book is going to use around that is about, um, free energy reduction, like free energy minimization, which is just to say in some really coarse grain, not completely correct way that like, it's just taking a lot of choices off of the table, a lot of actions that, um, would not help, would not cause the thing to persist or find more usefulness or, um, so, and, and the other two words here that I'm using, a pragmatic epistemic value is the epistemic value is kind of like going to find the, like foraging for the food. And then the, the pragmatic value is like, like actually eating, getting the food or something like that. Um, not exactly, but very rough, analogous sort of, I don't know if that's helpful, but

that's awesome. Thank you. Yeah. Many pieces here. So first you mentioned like the state space of possible moves, which is what is being suggested here with the first part of the sentence, the possibilities for control and adaptation are plentiful. So we can think about those in terms of affordances, which are relational capacities for action and the set of possible moves or affordances might be large. However, only like a vanishingly small subset of them might be useful. And then part of the challenge that's pointed to is that, um, there's a recursive nature of the cycle. So it's not like we can just say, well, you know, pawn to E4 is just simply a good move because there's probably situations where that like wins you the game. And there's probably situations where that loses the game. So within the utility defined by like a game, there's already immense challenges in exploring these like branching trees of outcomes and just evaluating them under any metric period. Um, so how can we evaluate perception and action and planning? The idea that's going to come into play is action and planning as divergence minimization. This can be contrasted with the notion of perception, cognition, action, planning, et cetera, as reward maximization. Reward maximization is equivalent to pragmatic value maximization. Utility optimization is pragmatic value. Um,

and we're going to see this reflected soon in some of the equations to come in the next chapters. But for chapter one, we're still addressing this like, um, qualitatively. Pragmatic value is like the task, you know, time on task. Um, if, if, if the pragmatism in the mountain climbing example is climbing the mountain, then pragmatic value is like every inch you take up the mountain. Epistemic value would be reducing your uncertainty about how to achieve pragmatic value. And so in that way, the free energy functional, which encompasses both pragmatic and epistemic value allows a more adept navigation of explore and exploit because under a reward maximization framework, it's very easy to like, you know, turn the speaker up to 11, set the phaser to stun and pursue local pragmatic value. However, all of these second order heuristics need to come into play to reintroduce like novelty bonuses, curiosity as a second level onto this, um, reward first pragmatic value first foundation. And in contrast, the imperatives for action and perception in active inference are going to be built around not reward maximization, but free energy minimization or relative free energy reduction, as mentioned, and that's going to be composed of, there's going to be multiple ways to break that down, what the free energy functional is, but it is going to often come down to, um, a pragmatic element in an epistemic element. Um, the pragmatic element is going to look like, again, we're, we're just beginning to come into these sections. The pragmatic value is going to look like a reduction of divergence between preferences and observations. So instead of saying, well, the most rewarding body temperature for me is 37 and I'm maximizing reward by staying at 37, that's a reward maximization perspective on temperature homeostasis. A divergence minimization perspective, surprise minimization perspective on body temperature homeostasis is I expect and prefer to find myself at 37. And I'm going to undertake policy to reduce my surprise about finding myself at 37. So when I'm at 37, I'll be minimally surprised. And as I get further, I will be more and more surprised. And then I will take pragmatic action to climb that hill as well as epistemic actions to find out how to climb that hill. But we're going to be getting back to 37 by reducing the divergence between preferences and outcomes rather than getting to 37 by like, like, hashtag winning and trying to maximize reward, which is the only fundamental, um, ontology or reward based learning. So that's how many people have in that, um, basal relationship between pragmatic and epistemic value and active inference. That's where many people have found space to explore like curiosity and novelty. And a lot of other things will open up once we first start to explore how pragmatic and epistemic value are like connected and as co-equal in the imperatives for active inference agents. And then there'll be even more fun things to come. Like again, when we realize that pragmatic value isn't maximizing rewards simply, it's reducing the divergence relative to expectations. So it's like, again, not, um, we're playing chess and we're trying to maximize our points and minimize their points, but we have an optimistic trajectory or set of expectations, and then we're going to reduce the divergence such that that is realized. So what determines useful? As with many other questions, there's like a limited answer that's so limited that it may not feel like an answer,

which is like, again, if we're balancing a plate on a stick, we may define utility as keeping the plate up.

If we were climbing a mountain, we may define utility in that limited dimension as getting to the top of the mountain.

That may feel like too narrow because we might think, well, like, but the person balancing the plate, I mean, they have to have a salary and they have to breathe and have blood sugar and stuff like that. So the limited answer

is like sort of just staying within the bounds of our model and the dimensions that we're considering.

And then part of this iterative modeling process, which especially starts to be introduced in the second half of the book, but also is reflected by people's experience, um, in other modeling areas, including qualitative modeling areas. Like, so then it's okay. So now we have the plate balancer and their blood sugar.

And then are we still missing something relative to the real world? Are we as modelers still being surprised by what happens now? We need to consider something else. So that's like the limited answer, which is utility is situational and dimensional in a narrow sense.

And then in perhaps a slightly more like miso or macro sense utility, um, or at least like, um, a cousin or a proxy of utility is survival and persistence because without survival and persistence, you're not going to be around to even have that debate about what's useful.

So it is kind of like a prerequisite or an enabling feature of any dimensional or like sort of one aspect.

You know, if you want to be in the game to, to realize any specific kind of utility,

being in the game is like that meta utility, which Brock pointed towards with like different, um,

behaviors and phenomena ranging from epistemic foraging to, um,

niche construction and like building a house.

Any other thoughts on utility? What determines useful?

Okay.

This looks like a great

long question. So I'm going to just bring it down so we can kind of break it into sections.

Um, can anyone describe the neat and scruffy distinction?

It's on page five.

Ali, yes.

It's again related to, uh, the high road and the low road. I mean, uh, they've described the high road as the needs perspective or the needs approach and, uh, the low road as the scruffy one, uh, probably because high road begins with some, uh, some established, uh, postulates and, uh, let's say, uh, without going into a bottom-up approach and beginning with, uh, the fundamental, uh, principles and then developing, uh, um, the propositions, uh, from there, but beginning, uh, with established propositions and then, uh, somehow

chopping it up or somehow going into a much more granular level in terms of its fundamental constituents.

thank you excellent i i um find this paragraph to be a very nice point which is

we can think of two extreme ends of a um kind of like research program or epistemic ecosystem

one would be every single neural process you know acetylcholine synapses neuro hormones being

circulated in this part every single physiological piece could have its own theory i mean they could have their own department they could have their own university um and same for every cognitive mechanism we could have a separate bespoke model with no interoperability at all there's no reason light has to between um attention cognition perception action social interaction communication anticipation planning we could have bespoke theories and somebody might suggest that that is like preferable in their eye whether for utility or for aesthetic reasons in their words that would lead to a proliferation of theories in fields such as the above with little hope for their unification another perspective is that despite their diverse manifestations the central aspects of behavior cognition and adaptation in living organisms are amenable to coherent explanation from first principles so in other words could we start with first principles you can explain or explore what those are but can we start with first principles and then see different biological processes and different cognitive mechanisms

as being under a common umbrella or under a common ontology or addressable by a shared model and then to come back to the question um because this is a nice key point of this uh chapter it's uh alluding to roger shank introducing this neat versus scruffy distinction and then as elise suggested this does map to the high road and a low road which um are mapped out in figure 1.2 and so the neats are like the lumpers

they're finding unity in heterogeneity and then the scruffies are like embracing the heterogeneity by focusing on details that demand dedicated explanations

i mean how can we have a model of all synapses when the acetylcholine synapses are so different than the dopamine synapses

and then we'll need a third model or a third paper to describe when there's both acetylcholine and dopamine at the same synapse

so that's like the scruffy bottom-up embracing the heterogeneity of nature and then the neat is coming from a top down or from the high road and finding unification across that diversification brock um sorry i just read this whole question and i don't want to kind of uh blow past um getting into the details of each but i want to try to maybe from where you started there kind of uh yeah thread uh the needle there through to the end of what what's kind of being asked here is that well he's he's the in the context of these you know uh bottom up kind of less ex more explanatory but less meaningful and the top down kind of more meaningful but maybe in some cases less explanatory or something like this um like how do we look at active inference or where does it sit in relation to all these other um theories and models of um mind body cognition etc and specifically in in friston's um essay uh the desert landscape here um he's got a quote about fep kind of subsuming all other theories as not some prediction but just that as they kind of take one at a time here they have kind of just happened to be commensurate and be describable in some way here by fep and so is that kind of like meaningless um

i lost you brock i'm not sure if that was me

hmm

hmm

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Well, sorry about this.

Not sure what is literally happening with my internet.

loading, loading, loading, loading, loading, loading, loading, loading, loading,

loading, loading, loading, loading, loading, loading, loading, loading, loading,

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Can you copy paste that risk?

That's a good response.

And I think that's...

I've appended the PDF that I've uploaded.

The...

Okay.

I was just gonna say that the desert landscape being left there is kind of like a new niche that looks maybe barren today, but if we continue, maybe it's not so barren.

Maybe there are adjacent things that are... that emerge from that niche being constructed,

that cultural lens being used that we look out and see something different that we weren't expecting.

Sorry about my strange absence.

I think...

I think...

Yeah, Brock.

I think...

Yeah, Brock.

Yeah, I can't hear you.

I think I can hear you.

Daniel, John, you are talking over her, so I don't know if you...

Yeah, I can't hear.

I'm gonna reload.

In a more simple frame from my perspective, which is not as advanced as anyone here, in the discussion, but it's more from exactly the descriptions you've been or what you've been talking about, has made it a bit more clear in a way.

I'm not sure about the high road and the low road, sort of bottom up, bottom down thing, but there's a lot of theories about everything.

But what came through then to me was that with FEP and active inference, it's a bit like it hasn't been fully explored, like it's...

that it's something where we're actually investigating ourselves so we don't actually know what we're going to come up with.

A bit like the sort of elephant with a whole lot of people, blind people trying to work out what the elephant is, and not actually getting a grasp of the whole thing.

And...

which makes it something that's going to...

that's still, as someone said, still evolving, still actually developing into whatever it will be eventually.

Yeah, which is helpful for my question in terms of if it's...

if it's going to consume every other theory, makes it a little bit too grand and a little bit too much, I think.

I don't know if I'm making sense, but it's quite a complex point.

Yeah, I think that that is...

that's... that is a very common response, and it's still...

I mean...

when you say you're not as advanced as...

I mean, we're just using words we've learned over the last two months, maybe, some of us, speaking personally.

So...

but...

you know, it's...

again, there's this question, like,

well, if it describes everything,

or, you know, and it's unfalsifiable, then what...

what does that mean?

You know, it's...

that's...

how is that helpful, or what is the use there?

I think...

Yeah.

What... what you keyed in on this sort of evolving that we don't know quite yet,

and how Ali...

called attention to that response from Carl,

of a kind of cultural lens is being developed, like, that...

that is...

kind of like niche construction,

that is...

that is us figuring out what is the extent of this niche.

Where does FAP apply?

Where does active inference apply, like, really well?

Yeah.

And then where does...

Daniel always uses linear regression, because it's so easy to grasp at.

Like, there's many cases where, you know, maybe active inference,

maybe some magical process is happening, but...

Wow, linear regression works really well, you know?

So...

Maybe...

you know, active inference applies, but...

it's a lot of work, and linear regression works really good, right?

Um...

So there are certainly...

I would expect, you know, boundaries...

to where the pragmatic use...

like, the usefulness of it is not.

Um...

it doesn't, um...

continue.

There's some...

Yeah.

Drop-off...

Yeah, so...

we don't know what that is precisely yet.

I certainly don't.

I don't know if somebody else does, but...

Um...

It's...

it's a question that's being explored, and I don't...

Yeah, I think we just...

we'll figure that out...

um...

over time, through exploring and finding where that desert,

you know, ends, kind of, or...

slowly turns back into an oasis for us to forage in.

Hmm.

Nice.

Thank you for this, um...

discussion.

I...

I do love the linear regression.

I mean, imagine if you said, really, numbers?

You think you're gonna be able to describe apples and oranges...

with numbers?

Or height and weight?

I mean, could two things be more different than time and height and weight?

You really think a linear regression is gonna cut through that?

Um...

So...

sometimes...

modeling and statistics overall...

can feel like they're encompassing a lot.

And then especially when there's so much philosophy...

and reflection...

and, like, relational thinking...

in the mix...

it truly can...

converge on what feels like a theory of everything.

And, um...

one great video is, um...

from a channel called Theories of Everything...

by Kurt Jaimungel.

And, um...

it's with Karl Fristen...

there's several with Karl.

And so, that is tackled head-on.

And I don't exactly...

um...

recall...

whether he goes this direction, but...

the...

um...

quintessentially...

Fristonian...

twist...

is...

yeah, we are talking about...

everything.

Everything...

that for you...

is in this sort of, like, process-object-dialectic.

It is realizing a non-equilibrium steady-state.

Otherwise, you wouldn't call it a thing.

whatever it is that separates that thing...

from what is not that thing...

is like a blanket.

And if you're gonna say it's a fuzzy blanket, well...

there's some...

blanket index and there's a space for that discussion as well.

Um...

but when we're talking about things...

as they are...

and or...

as we model them...

we can put them into a framework of...

particular things.

And that's the free energy principle for a particular physics...

also known as the Bayesian mechanics.

So...

it...

it does take a slightly different approach...

and then...

what will...

active inference converge upon...

or move between?

these are...

definitely...

open and developing...

questions.

and also...

I like what...

like...

about the desert and the oasis...

because...

the desert is...

alive as well.

But...

this is a reference to this...

um...

to Quine's desert landscape.

Um...

and some of that work...

which has been referenced.

So...

very...

interesting.

And then there's also like a sort of interpersonal...

approach...

slash...

like...

is...

this presented as...

are you presenting...

active from the high road?

Well...

all things need to resist dissipation...

so that's why we're using...

non-equilibrium steady states...

to describe how things exist...

and those kinds of systems...

have to be doing something...

that either is active inference...

or that we can describe as active inference.

And starting with an imperative...

of persistence...

or...

coming from the low road...

and thinking about...

instances...

and how we can...

model what they're doing...

using Bayesian methods...

many times...

people will find resonance...

with either or both of those roads...

and...

there's a huge amount of...

ontology mapping...

availabilities...

for people to connect...

different theories to each other.

and also with...

and...

next month...

with the Bayesian mechanics...

discussion...

there...

we'll...

we'll tackle this Bayesian mechanics...

head on...

and...

so it's just...

I mean...

year after year...

and month after month...

there's a lot to...

learn and understand here...

and sometimes...

the philosophers...

outrun...

and anticipate...

where things may...

go...

and then...

sometimes...

the writing...

and the genre...

can feel like...

something has already been achieved...

like...

there's already been a synthesis...

of an activism...

and active inference...

it can feel like...

something that is being...

doubted...

is like a conclusive...

statement...

that it cannot be addressed...

Markov blankets...

cannot address...
physical interfaces...

sometimes...
lifted from the modeling...
there can be...
writing...
and conversation...
that might seem like...
it is like...
more or less confident...
about...
the status...
or future...

of this area...
which is like...
what...
motivates...
this kind of like...
reading between the lines...
and between the citations...
and also...
a deflationary...
modeling...
approach...
where...
like...
um...
um...
um...
um...
to....

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

I suppose that's where I've got to at the moment at this point.

Yeah.

And just, you know, the overriding, I'm not sure that's, I wouldn't worry about it overriding theories or, you know, things that kind of already are useful.

Okay.

Right.

It's, you know, relativity doesn't make, you know, Newton's laws irrelevant.

Right.

So, where that boundary is specifically is a difficult question right now.

Yeah.

But I think the vast majority of theory models of these cognition are not, that are, you know, historically been useful are not being thrown out.

right now they're just being, we're trying to connect them.

Right.

So.

I would also just add to that.

There's like the instrumentalist debate versus the realist debate.

And I think that, you know, the active inference is not necessarily to be taken as this is the way that it is more that it's a tool that we can use in our, in our toolbox.

And so I don't know that it invalidates any other theory, but it is just another, another tool in the toolbox, right?

Another thing that we can use to model things.

And definitely like, there are some things that active inference, or at least as I see it, it doesn't explain yet, like, like self-sacrificing behavior, right?

Like, like it, it doesn't necessarily explain that yet, but maybe if you look across different temporal skills, that that's like possible or like, yeah, like negative harm and harmful, like self-harm.

So active inference, I don't think maybe we'll ever successfully model that.

But also like one more thing I just want to mention.

I don't think anyone here, or I don't want to speak for everyone, but I don't think anyone in the classroom today feels like they're at the top of that learning curve.

So, like I've been studying active inference for like years already, and I still feel like maybe I'm in the long tail end of the trails on the way up the mountain.

So just, just, just know that it's definitely a process.

Yeah, I think I've realized that.

I mean, I've been reading about it for a few years, but, and in relationship to, to, to what I do as a, I'm an Alexander Technique teacher.

So I'm looking at it from that perspective and just seeing how it fits in.

And so it comes up against other motor control theories.

And, you know, I mean, to, to, to say something really simple is the idea that we, that we don't have motor commands.

We have predictions.

Just the difference between that.

And if I was going to put forward active inference to my colleagues, then, then how do I substantiate that?

Or how do I explain that?

Is, is just a question in my mind, I suppose.

Do I say that, you know, the motor command thing's out the window and it's all prediction, or do I just say, this is a possibility?

This is one idea.

Yeah.

So that's a simple point.

Great.

Great.

Point.

Is that Daniel?

Yeah.

Any, oh, funny.

I can, I can hear you though.

So it's on the recording.

Anyone else have any last comments or questions?

Nice.

It's, it's great.

Yes.

Go ahead.

Yeah.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

So I exchanged some emails with Thomas Parr.

Because I went through the book and I, you know, I noticed some things that may, may have been some typos or some, or asking some, some questions.

And so I, I compiled the document on Overleaf with these points and questions.

And he was very kind to answer and to provide answers and about also some clarification of some equations.

So I'm, I sent him an email and asked him if he was okay to share it with the group, to share the document with the group.

And he hasn't answered yet, but I'm pretty sure he will consent.

And then, you know, I know where, where would be the best place to put this document.

Great.

Thank you.

Thank you.

So in this errata section, here's where we've, um, have, have, um, a handful of, uh, typos and ambiguities and I'll make a sub page.

Okay.

And, um, you'll have full control.

So just anything you want to do in this sub page, just do it.

Okay.

Um, all right.

Well, great time.

Sorry about my strange internet just disrupts beyond belief, but we did it.

I'll, uh, upload this soon and, uh, enjoy reading through chapter two.

Please add any question, the most basic and the most confirmatory to the most speculative.

Every question will be really amazing to see for chapter two.

So thanks everyone.

And see you soon.

Thank you.

Thank you.